
EECS 16A Designing Information Devices and Systems I

Homework 1

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any hand-written answers should be scanned) as well as your IPython notebook saved as a PDF.

If you do not attach a PDF “printout” of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

1. Reading Assignment

For this homework, please read Notes 1A, 1B and 2A. The notes will provide an overview of systems of linear equations, vectors, and matrices. You are always welcome and encouraged to read beyond this as well, in particular, a quick look at Note 3 will help you.

- Describe how Gaussian Elimination can help you understand if there are no solutions to a particular system of equations?
- What about a unique solution?
- Does a row of zeros always mean there are infinite solutions?

2. Gaussian Elimination

- (a) In this problem we will investigate the relationship between Gaussian elimination and the geometric interpretation of linear equations. You are welcome to draw plots by hand or using software. Please be sure to label your equations with a legend on the plot.

- i. Draw the following set of linear equations in the x - y plane. If the lines intersect, write down the point or points of intersection.

$$x + 2y = 4 \tag{1}$$

$$2x - 4y = 4 \tag{2}$$

$$3x - 2y = 8 \tag{3}$$

- ii. Write the above set of linear equations in augmented matrix form and do the first step of Gaussian elimination to eliminate the x variable from equation 2. Now, the second row of the augmented matrix has changed. Plot the corresponding new equation created in this step on the same graph as above. What do you notice about the new line you draw?
- iii. Complete all of the steps of Gaussian elimination including back substitution. Now plot the new equations represented by the rows of the augmented matrix in the last step (after completing back substitution) on the same graph as above. What do you notice about the new line you draw?
- (b) Write the following set of linear equations in augmented matrix form and use Gaussian elimination to determine if there are no solutions, infinite solutions, or a unique solution. If any solutions exist, determine what they are. You may do this problem by hand or use a computer. We encourage you to try it by hand to ensure you understand Gaussian elimination.

$$\begin{aligned}
 x + 2y + 5z &= 3 \\
 x + 12y + 6z &= 1 \\
 2y + z &= 4 \\
 3x + 16y + 16z &= 7
 \end{aligned}$$

(c) Consider the following system of equations:

$$\begin{aligned}
 x + 2y + 5z &= 6 \\
 3x + 9y + 6z &= 3
 \end{aligned}$$

You are given a set S of candidate solutions,

$$S = \left\{ \vec{v} \mid \vec{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ -5 \\ 0 \end{bmatrix} + \begin{bmatrix} -11 \\ 3 \\ 1 \end{bmatrix} t, \quad t \in \mathbb{R} \right\}$$

[Note: $\in$ is a mathematical symbol meaning "is an element of".]

This vector notation can be expressed in terms of its components:

$$\vec{v} = \begin{bmatrix} 16 - 11t \\ -5 + 3t \\ t \end{bmatrix} \quad \text{means} \quad \begin{aligned} x &= 16 - 11t \\ y &= -5 + 3t \\ z &= t \end{aligned}$$

Show, by substitution, that any $\vec{v} \in S$ is a solution to the system of equations given above. Note that this means that the candidate solution must satisfy the system of equations for all $t \in \mathbb{R}$.

(d) Consider the following system:

$$\begin{aligned}
 4x + 4y + 4z + w + v &= 1 \\
 x + y + 2z + 4w + v &= 2 \\
 5x + 5y + 5z + w + v &= 0
 \end{aligned}$$

If you were to write the above equations in augmented matrix form and use Gaussian elimination to solve the system, you would get the following (for extra practice, you can try and do this yourself):

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 3 & 16 \\ 0 & 0 & 1 & 0 & -3 & -17 \\ 0 & 0 & 0 & 1 & 1 & 5 \end{array} \right]$$

How many variables are free variables? Determine the solutions to the set of equations.

3. Filtering Out The Troll

Learning Goal: The goal of this problem is to represent a practical scenario using a simple model of directional microphones. You will tackle the problem of sound reconstruction through solving a system of linear equations.

You attended a very important public speech and recorded it using a recording device that had two directional microphones¹. However, there was a person in the audience who was trolling around, adding interference to the recording. When you went back home to listen to the recording, you realized that the two recordings were dominated by the troll's interference and you could not hear the speech. Fortunately, since your recording device contained two microphones, you realized there is a way to combine the two individual microphone recordings so that the troll's interference is removed. You remembered the locations of the speaker and the troll and created the diagram shown in Figure 1. You (and your two microphones) are located at the origin.

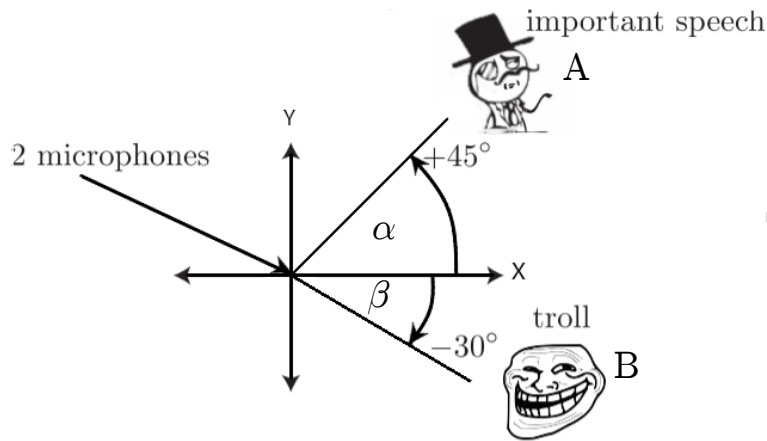


Figure 1: Locations of the speaker and the troll.

Each directional microphone records signals differently based on their angles of arrival. The first microphone weights (multiplies) a signal, coming from an angle θ with respect to the x-axis, by the factor $f_1(\theta) = \cos(\theta)$. If two signals are simultaneously playing (as is the case with the speech and the troll interference), then a linear combination (i.e. a weighted sum) of both the signals is recorded, each weighted by the respective $f_1(\theta)$ for their angles. The second microphone weights a signal, coming from an angle θ with respect to the x-axis, by the factor $f_2(\theta) = \sin(\theta)$. Again, if there are two simultaneous signals, then the second microphone also records the weighted sum of both the signals.

For example, an audio source that lies on the x axis will be recorded by the first microphone with weight equal to 1 (since $\cos(0) = 1$), but will not be recorded up by the second microphone (since $\sin(0) = 0$). Note that the weights can also be negative.

Let us represent the speech sample at a particular time-instant by the variable a and the interference caused by the troll at the same time-instant by the variable b . Remember, we do not know either a or b . The recording of the first microphone at that time instant is given by m_1 :

$$m_1 = f_1(\alpha) \cdot a + f_1(\beta) \cdot b,$$

and the second microphone recorded the signal

$$m_2 = f_2(\alpha) \cdot a + f_2(\beta) \cdot b.$$

¹If you want to learn more about directional microphones, click on this [link](#)

where α and β are the angles at which the public speaker A and the troll B respectively are located with respect to the x-axis, and vectors a and b are the audio signals produced by the public speaker A and the troll B respectively.

(As a side note, we could represent the entire speech with a vector $\vec{a}$ by stacking all the speech samples on top of each other, and this is what we would typically do in a real-world speech processing example. However, here we consider just one time instant of the speech for simplicity.)

- Plug in the values of α and β to write the recordings of the two microphones m_1 and m_2 as a linear combination (i.e. a weighted sum) of a and b .
- Solve the system you wrote out on the earlier part to recover the important speech a , as a weighted combination of m_1 and m_2 . In other words, write $a = u \cdot m_1 + v \cdot m_2$ (where u and v are scalars). What are the values of u and v ?
- Partial IPython code can be found in `probl.ipynb`, which you can access through the datahub link associated with this assignment on the course website. Complete the code to get the signal of the important speech. Write out what the speaker says. (Optional: Where is the speech taken from?)

Note: You may have noticed that the recordings of the two microphones sound remarkably similar. This means that you could recover the real speech from two “trolled” recordings that sound almost identical! Leave out the fact that the recordings are actually different, and have some fun with your friends who aren’t lucky enough to be taking EECS16A.

4. Anish’s Optimal Tea

Learning Goal: *Recognize a problem that can be cast as a system of linear equations.*

Anish’s Optimal Tea has a unique way of serving its customers. To ensure the best customer experience, each customer gets a combination drink personalized to their tastes. Anish knows that a lot of customers don’t know what they want, so when customers walk up to the counter, they are asked to taste four standard combination drinks that each contain a different mixture of the available pure teas.

Each combination drink (Classic, Roasted, Mountain, and Okinawa) is made of a mixture of pure teas (Black, Oolong, Green, and Earl Grey), with the total amount of pure tea in each combination drink always the same, and equal to one cup. The table below shows the quantity of each pure tea contained in each of the four standard combination drinks.

Tea [cups]	Classic	Roasted	Mountain	Okinawa
Black	$\frac{1}{3}$	$\frac{1}{3}$	0	$\frac{2}{3}$
Oolong	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{2}{5}$	$\frac{1}{3}$
Green	0	$\frac{1}{3}$	$\frac{3}{5}$	0
Earl Grey	$\frac{1}{3}$	0	0	0

Initially, the customer’s ratings for each of the pure teas are unknown. Anish’s goal is to determine how much the customer likes each of the pure teas, so that an optimal combination drink can then be made. By letting the customer taste and score each of the four standard combination drinks, Anish can use linear algebra to determine the customer’s initially unknown ratings for each of the pure teas. After a customer gives a score (all of the scores are real numbers) for each of the four standard combination drinks, Anish

then calculates how much the customer likes each pure tea and mixes up a special combination drink that will maximize the customer's score.

The score that a customer gives for a combination drink is a linear combination of the ratings of the constituent pure teas, based on their proportion. For example, if a customer's rating for black tea is 6 and oolong tea is 3, then the total score for the Okinawa Tea drink would be $6 \cdot \frac{2}{3} + 3 \cdot \frac{1}{3} = 5$ because Okinawa has $\frac{2}{3}$ black tea and $\frac{1}{3}$ Oolong tea.

Professor Ranade was thirsty after giving the first lecture, so Professor Ranade decided to take a drink break at Anish's Optimal Tea. Professor Ranade walked in and gave the following ratings:

Combination Drink	Score
Classic	7
Roasted	7
Mountain	$\frac{37}{5}$
Okinawa	$\frac{19}{3}$

- What were Professor Ranade's ratings for each tea? **Show your work for this problem, step by step. You may use a calculator to do algebra.**
- What mystery tea combination could Anish put in Professor Ranade's personalized drink to maximize the customer's score? If there is more than one correct answer, state that there are many answers, and give one such combination. What score would Professor Ranade give for the answer you wrote down? Assume the total amount of tea must be one cup.

5. Fountain Codes

Learning Goal: *Linear algebra shows up in many important engineering applications. Wireless communication and information theory heavily rely on principles of linear algebra. This problem illustrates some of the techniques used in wireless communication.*

Vivian wants to send a message to her friend Kanav. Vivian sends her message $\vec{m}$ across a wireless channel in the form of a transmission vector $\vec{v}$. Kanav receives a vector of symbols denoted as $\vec{r}$.

Vivian knows some of the symbols in the transmission vector that she sends may be corrupted, so she needs a way to protect her message from the corruptions. (Transmission corruptions occur commonly in real wireless communication systems, for instance, when your laptop connects to a WiFi router, or your cellphone connects to the nearest cell tower.) Ideally, Vivian can come up with a transmission vector such that if some of the symbols get corrupted, Kanav can still figure out what Vivian is trying to say!

One way to accomplish this goal is to use fountain codes, which are part of a broader family of codes called error correcting codes. The basic principle is that instead of sending the exact message, Vivian sends a modified longer version of the message so that even if some parts are corrupted Kanav can recover what she meant. Fountain codes are based on principles of linear algebra, and were actually developed right here at Berkeley! The company that commercialized them, Digital Fountain, (started by a [Berkeley grad, Mike Luby](#)), was later acquired by Qualcomm. In this problem, we will explore some of the underlying principles that make fountain codes work in a very simplified setting.

The message that Vivian wants to send to Kanav are the three numbers a , b , and c . The message vector representing these numbers is $\vec{m} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$. Figure 2 shows how Vivian's message is encoded in a transmission vector and how Kanav's received vector may have some corrupted symbols.

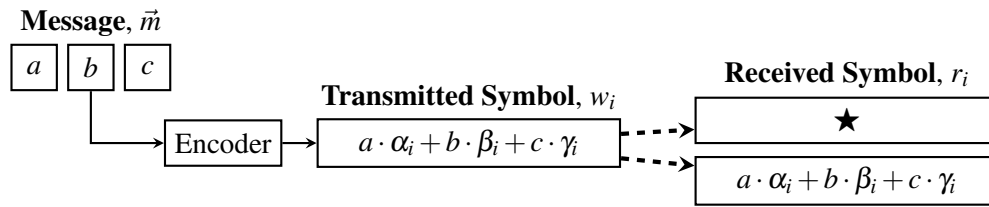


Figure 2: Each symbol in a transmission vector $\vec{w}$ is a linear combination of a , b , and c . Each transmitted symbol, w_i , is either received exactly as it was sent, or it is corrupted. A corrupted symbol is denoted by $\star$. The i th row of a symbol generating matrix G determines the values of α_i , β_i , and γ_i .

[Note: You may find it helpful to familiarize yourself with some Greek letters, to be able to effectively refer to them in this course! See [Greek Alphabet](#).]

- (a) Since Vivian has three numbers she wishes to send, she could transmit six symbols in her transmission vector for redundancy. This transmission strategy is called the **repetition code**.

If Vivian uses the repetition code, her transmission vector is $\vec{w} = \begin{bmatrix} a \\ b \\ c \\ a \\ b \\ c \end{bmatrix}$.

As depicted in Figure 2, the received vector may have corrupted symbols. For example, suppose only

the first symbol was corrupted, then Kanav would receive the vector $\vec{r} = \begin{bmatrix} \star \\ b \\ c \\ a \\ b \\ c \end{bmatrix}$, where the $\star$ symbol

represents a corrupted symbol.

Using the repetition code scheme, give an example of a received vector $\vec{r}$ with only two corrupted symbols such that a is unrecoverable but b and c are still recoverable.

- (b) Vivian can generate $\vec{w}$ by multiplying her message $\vec{m} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ by a matrix. Write a matrix-vector multiplication that Vivian can use to generate $\vec{w}$ according to the repetition code scheme. Specifically,

find a “generating” matrix G_R such that $G_R \vec{m} = \vec{w}$, where $\vec{w} = \begin{bmatrix} a \\ b \\ c \\ a \\ b \\ c \end{bmatrix}$.

- (c) Instead of a repetition code, it is also possible to use other codes (e.g. fountain codes). Vivian and Kanav can choose any symbol generating matrix, as long as they agree upon it in advance. Each different matrix represents a different “code.” Vivian’s TA recommends using the symbol generating matrix G_F :

$$G_F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Vivian then uses the symbol generating matrix G_F to produce a new transmission vector: $G_F \vec{m} = \vec{w}$.

Suppose Kanav receives the vector $\vec{r} = \begin{bmatrix} 7 \\ \star \\ \star \\ 3 \\ 4 \\ \star \\ \star \end{bmatrix}$, which is a corrupted version of $\vec{w}$.

Write a system of linear equations that Kanav can use to recover the message vector $\vec{m}$. Solve it to recover the three numbers that Vivian sent.

Hint: Consider the rows of G_F that correspond to the uncorrupted symbols in $\vec{r}$.

- (d) We explore one example case where Vivian and Kanav agree to use G_F (i.e. $G_F \vec{m} = \vec{w}$) and there are three corruptions, so Kanav receives four uncorrupted symbols.

Suppose Kanav receives $\vec{r} = \begin{bmatrix} 1 \\ \star \\ 3 \\ \star \\ 4 \\ \star \\ 9 \end{bmatrix}$.

Can you determine the message $\vec{m}$ that Vivian sent? If so, determine $\vec{m}$.

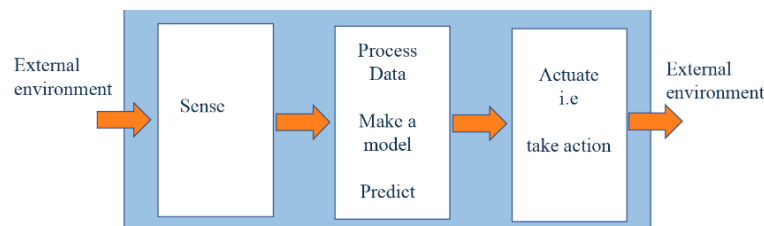
Note: it can be shown that receiving *any* four uncorrupted symbols when Vivian is using G_F is enough to recover Vivian's message. On the other hand, we showed in part (a) that receiving *any* four uncorrupted symbols using G_R does not guarantee we can recover Vivian's message. This is why, in practice, we would prefer to use the fountain code G_F instead of the repetition code G_R — it is a more reliable way to send messages.

6. Group Problem: Explore an Interesting Application of EECS

This problem is meant to be solved in your study groups and start a discussion. The conversation you have is more important to us than the answers you write down to this problem. If you chose to not have a study group, you may do this on your own, or discuss with other students during homework party.

As a group, identify an exciting technology you would like to build. In this problem, you will go through the design process for this technology by identifying the required sensing, processing and actuation components. Some examples of technologies you could consider are the step counters on phones, GPS, self-driving cars, a robotic drummer accompanist, a fitbit, an advanced pacemaker, a hearing aid, touchscreens, an automated fruit picker that picks ripe fruit, augmented reality for emotional awareness, a car that can apologize to defuse road rage, a friend bot, a pollution cleaning-bot etc.

This task is intended to be a fun exploration, so responses for each part need not exceed a paragraph in length. You are also encouraged to come to office hours to connect with your classmates and share ideas!



- (a) What kind of information does your technology need to collect from the external environment? How is this collected? Are any sensors used? If so, what physical quantities do they measure?
- (b) How do you propose to process the data obtained from the sensor measurements? What would you need to model? Would you have to make any predictions from the data? What techniques would you use?
- (c) What actions would you want to take in the real world using your prediction or processed data? Describe the actuation mechanisms, if applicable, in this technology.
- (d) What are the potential social implications of this technology? Good? Bad? Transformative? Redistributive?

7. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.